

IntelliSafe

Real-Time Event-Driven Industrial Safety Monitoring using Computer Vision

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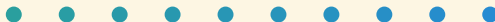
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Presentation Overview

Sections

1. Problem Statement
2. Proposed Solution
3. Detection Models
4. Compliance Feature
5. Alert System
6. Technology Stack
7. Project Timeline
8. Deliverables
9. Demo Plan
10. Summary



The Problem: Industrial Safety Challenges

Current Reality:

- **2.3 million** workers die annually from work-related accidents (ILO)
- **340 million** occupational accidents per year worldwide
- **70%** of accidents involve PPE non-compliance

Key Issues:

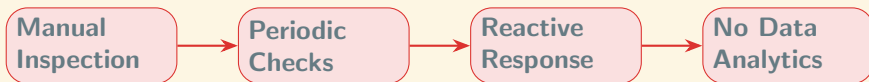
- Manual safety inspections are sporadic
- Human supervisors cannot monitor 24/7
- Violations go undetected until accidents occur
- No real-time compliance tracking

Impact

- Loss of life
- Permanent disabilities
- Legal liabilities
- Production downtime
- Insurance costs

Why Current Solutions Fail

Current Approach - INEFFECTIVE



Problems:

- Coverage gaps between inspections
- Human error and fatigue
- No evidence for incidents
- Subjective assessments

Need:

- Continuous monitoring
- Automated detection
- Instant alerts
- Data-driven insights

Our Solution: AI-Powered Safety Monitoring

Real-Time Industrial Safety Monitoring System

Core Capabilities

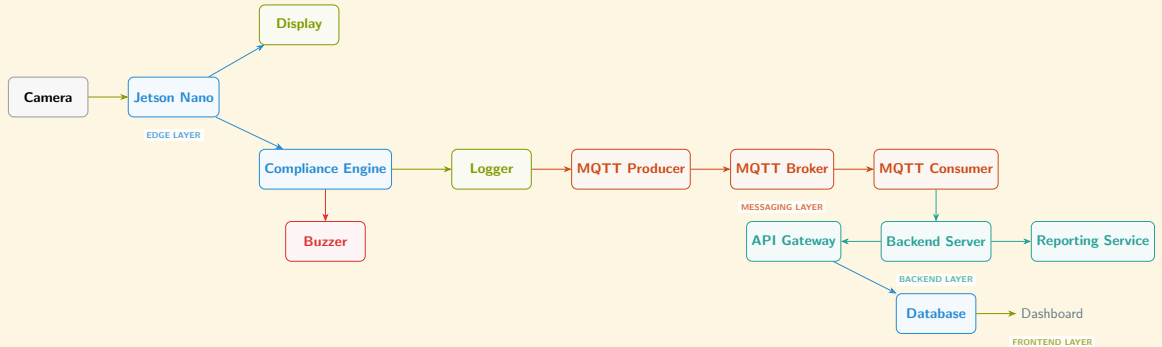
- Real-time PPE detection using AI
- Automated compliance scoring
- Instant violation alerts
- Evidence capture with images
- Historical analytics & reports

Key Features

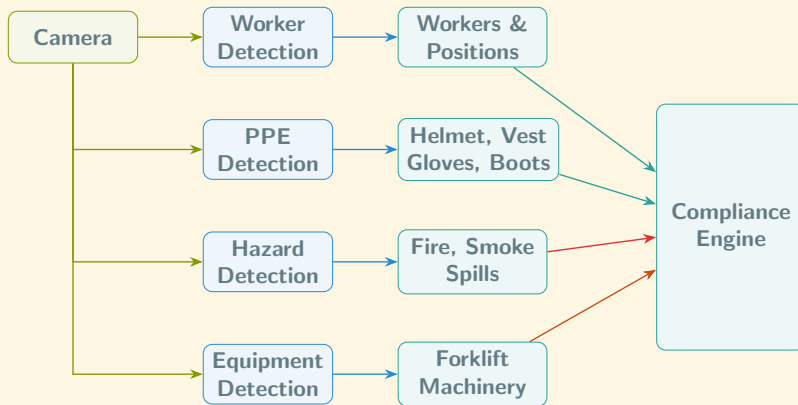
- Edge processing (low latency)
- Multi-model detection pipeline
- Event-driven architecture
- Web dashboard monitoring
- PDF report generation

Transform reactive safety management into proactive, AI-driven compliance monitoring

System Architecture Overview



Four Detection Models Architecture



Model Details and Detection Classes

Model	Detects	Classes	Purpose
1	Workers	worker	Count, track, trigger PPE checks
2	PPE	helmet, vest, gloves, goggles, boots	Verify safety equipment compliance
3	Hazards	fire, smoke, spill	Detect environmental dangers
4	Equipment	forklift, machinery	Ensure proximity-based safety

Model Architecture

- YOLOv8n (Nano variant)
- Optimized for edge devices
- 640×480 resolution
- Real-time (>15 FPS)

10 Detection Classes Total

- 1 worker class
- 5 PPE classes
- 3 hazard classes
- 1 equipment class

Compliance Engine: How It Works

PPE Association Logic:

- Match PPE to worker using bounding box overlap
- Calculate Intersection over Union (IoU)
- Associate PPE if $\text{IoU} > 0.3$

Compliance Rules:

Condition	Status
All PPE present	COMPLIANT
Missing helmet	VIOLATION
Missing vest	VIOLATION
Missing ≥ 2 PPE	HIGH PRIORITY
Fire/Smoke detected	CRITICAL
Near equipment	WARNING

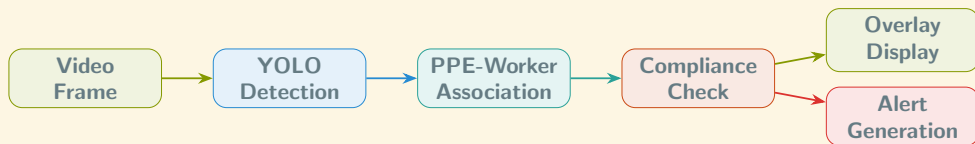
Compliance Score Formula

$$\text{Score} = \frac{\text{Safe Frames}}{\text{Total Frames}} \times 100$$

Score Interpretation

- $\geq 90\%$: Excellent
- 80 – 89%: Good
- 70 – 79%: Needs Improvement
- $< 70\%$: Critical

Real-Time Compliance Visualization



Overlay Display Shows:

- Green box = Compliant worker
- Red box = Non-compliant worker
- Missing PPE labels
- Live compliance percentage

Evidence Capture:

- Timestamped images saved
- JSON logs with metadata
- Confidence scores recorded
- Violation type logged

Multi-Level Alert System

Event	Priority	Action	Response
Fire/Smoke Detected	CRITICAL	Immediate Alert	Evacuation warning
Worker Near Equipment	HIGH	Audio Warning	Proximity alert
Missing 2+ PPE Items	HIGH	Supervisor Alert	Immediate action
No Helmet	MEDIUM	Log + Dashboard	Verbal warning
No Vest	MEDIUM	Log + Dashboard	Verbal warning
Compliance < 80%	MEDIUM	Generate Report	Review meeting

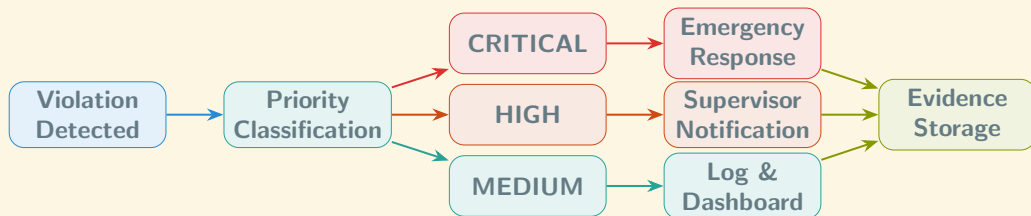
Alert Delivery Channels

- On-screen visual overlay
- Dashboard notifications
- Audio warnings (edge)
- Email alerts (backend)

Alert Data Captured

- Timestamp
- Violation type
- Confidence score
- Evidence image path

Alert Flow Architecture



All alerts include evidence capture (images + metadata) for compliance auditing and incident investigation

Technology Stack Overview

Edge Device (Jetson Nano)

OS	Ubuntu 18.04/JetPack
Language	Python 3.8+
AI Framework	Ultralytics YOLO
Vision	OpenCV 4.x
Messaging	Paho MQTT Client
Acceleration	CUDA, TensorRT

Hardware

Device	Jetson Nano 2GB
Storage	32GB microSD
Camera	USB/CSI Camera

Backend Server

Framework	FastAPI / Flask
Database	MongoDB / PostgreSQL
Broker	MQTT (Mosquitto)
Reports	ReportLab
Analytics	Pandas, Matplotlib

Frontend Dashboard

Framework	React.js / Vue.js
Charts	Chart.js / Recharts
Styling	Tailwind CSS

Sprint Breakdown Summary

Sprint	Focus	Key Deliverables
1	Setup & Data	Dev environment, annotated dataset (1000+ images)
2	Model Training	Trained YOLO models, TensorRT optimization
3	Edge Deployment	Jetson running detection, compliance engine
4	Messaging	MQTT event streaming, alert pipeline
5	Backend	FastAPI server, MongoDB database, consumer service
6	Dashboard	React dashboard with real-time updates
7	Reports	PDF generation, full system integration
8	Testing & Demo	Test cases passed, documentation, live demo

Project Deliverables

Software Deliverables

- Trained YOLO detection models
- Edge inference application
- Compliance engine module
- MQTT event streaming
- Backend API server
- Web dashboard
- PDF report generator

Documentation

- System design document
- API documentation
- User manual
- Installation guide
- Test report

Demo

- Live working system
- Video recording
- Presentation slides

Live Demo Execution Plan

① System Architecture Overview

- Explain edge-to-cloud data flow

② Compliant Scenario

- Worker wearing all PPE → Green bounding box

③ Violation Detection

- Remove helmet → Red box + alert generated

④ Compliance Score Update

- Show real-time score dropping

⑤ Evidence Capture

- Show saved violation image and JSON log

⑥ Dashboard Monitoring

- View alerts on web dashboard

⑦ Report Generation

- Generate and display PDF compliance report

Project Summary

Real-Time Industrial Safety Monitoring System

Edge AI • Event-Driven • Full-Stack

Technical Highlights:

- 4 YOLO detection models
- Real-time edge processing
- MQTT event streaming
- Cloud analytics & dashboards
- Automated compliance scoring

Impact:

- Continuous 24/7 monitoring
- Instant violation alerts
- Evidence-based compliance
- Data-driven safety insights
- Reduced accident risk

Timeline: 8 Months (32 Weeks) — **Platform:** Jetson Nano + Cloud

Thank You

Questions & Discussion

Real-Time Industrial Safety Monitoring System
COMSATS University Islamabad (Lahore Campus)